

PhysiNet-Operator: A Physics-Informed Neural Operator Framework for Spatiotemporal PDE Fields

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Abstract

Modeling physical systems governed by partial differential equations (PDEs) is essential across scientific and engineering domains, including fluid mechanics, wave propagation, climate dynamics, and materials science. Classical numerical solvers are accurate but computationally expensive, especially for long spatiotemporal rollouts, parameter sweeps, or real-time applications. Neural operators provide an emerging alternative by learning solution operators that map between infinite-dimensional function spaces in a resolution-invariant manner.

This paper introduces *PhysiNet-Operator*, a research-grade framework for learning operator mappings in physical systems using physics-informed neural operators, probabilistic generative modeling, and a synthetic PDE simulation gym built in JAX. The framework consists of four main components: (i) *PhysiGym*, a differentiable PDE environment supporting several canonical PDE families (2D wave, heat, advection–diffusion, and Gray–Scott reaction–diffusion); (ii) a hybrid neural operator architecture combining a multiscale encoder, Fourier Neural Operator (FNO) core, physics-informed residual blocks, and auxiliary PDE feature channels; (iii) a modular, HPC-friendly training pipeline for deterministic and probabilistic operator learning; and (iv) tools for visualization, uncertainty estimation, and spectral analysis.

We empirically evaluate PhysiNet-Operator on synthetic PDE benchmarks generated by PhysiGym, and discuss how it can be aligned with benchmark suites such as PDEBench. Results demonstrate accurate multi-step rollouts, stable long-horizon predictions, resolution generalization, and

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meaningful uncertainty estimates across multiple PDE families. This framework provides a unified platform for PDE modeling and serves as a foundation for future research in physics-informed machine learning, operator learning, and scientific AI.

Keywords: Neural operators, physics-informed learning, Fourier Neural Operator, PDE modeling, JAX, uncertainty estimation, generative models, scientific machine learning

1. Introduction

Many natural and engineered systems—including fluid flows, structural dynamics, electromagnetism, and geophysical processes—are governed by partial differential equations (PDEs). Numerical solvers based on finite differences, finite volumes, finite elements, or spectral methods remain the standard tools for simulating these systems. However, their computational cost grows rapidly with spatial resolution, time horizon, and parameter dimensionality, limiting their use in real-time control, high-throughput design, and large-scale uncertainty quantification.

The growing field of *scientific machine learning* (SciML) aims to complement classical solvers with data-driven surrogates and hybrid methods. A particularly promising direction is *operator learning*, where the goal is to learn mappings between function spaces rather than fixed-size vectors or tensors. Neural operators such as the Fourier Neural Operator (FNO) [1], DeepONet [2], and transformer-based operator models [6, 7] have been shown to generalize across spatial resolutions and parametrizations, making them attractive candidates for PDE surrogates.

Despite rapid progress, several practical gaps remain:

- the lack of unified, differentiable environments for generating diverse PDE datasets,
- limited integration of physics-informed structure into neural operator architectures,
- the need for uncertainty-aware predictions in safety-critical or decision-making contexts,
- and the absence of research-grade, extensible frameworks that tie together simulation, operator learning, and evaluation.

To address these challenges, we introduce *PhysiNet-Operator*, an integrated framework that combines:

1. **PhysiGym**, a JAX-based PDE simulation gym for generating synthetic spatiotemporal fields;
2. a **multiscale neural operator architecture** built around FNO, physics-informed residual blocks, and auxiliary PDE channels;
3. a **training pipeline** for deterministic and probabilistic operator learning with autoregressive rollouts;
4. and a **visualization and analysis suite** for inspecting fields, spectra, and uncertainty.

PhysiNet-Operator is designed to be resolution-invariant, physics-aligned, and extensible. While this paper focuses on canonical PDEs (wave, heat, advection–diffusion, and Gray–Scott), the framework is intended to be extended to more complex domains and integrated with benchmark suites such as PDEBench [11].

1.1. Contributions

The main contributions of this work are:

- We propose **PhysiNet-Operator**, a physics-informed neural operator framework that unifies differentiable PDE simulation, operator learning, and probabilistic modeling for spatiotemporal fields.
- We introduce **PhysiGym**, a JAX-based synthetic PDE environment supporting several canonical PDE families with randomized coefficients, multiresolution grids, and exportable datasets.
- We design a **hybrid neural operator architecture** that combines a multiscale encoder, FNO core, physics-inspired residual blocks, and optional probabilistic heads for uncertainty estimation.
- We develop a **modular training pipeline** that supports operator learning objectives, autoregressive rollouts, and probabilistic losses, implemented in JAX/XLA for efficient accelerator execution.
- We provide **qualitative and quantitative evaluations** on synthetic PDE tasks and discuss how the framework can be aligned with recent benchmarks and transformer-based neural operators.

2. Related Work

2.1. Numerical PDE Solvers

Classical numerical solvers based on finite differences, finite volumes, finite elements, and spectral methods remain highly effective for a wide variety of PDEs. They provide guarantees on stability and convergence for well-posed problems, but often at high computational cost. In large-scale or real-time settings, surrogate models and reduced-order methods are frequently employed to alleviate these costs.

2.2. Data-Driven PDE Surrogates

Deep learning methods have been applied to PDE surrogates using convolutional networks, U-Nets, and recurrent architectures. For instance, convolutional encoders-decoders and ConvLSTMs have been used for forecasting fluid flows and climate fields. However, these architectures are tied to fixed grid resolutions and often lack the inductive bias for operator learning across function spaces.

2.3. Neural Operators

Neural operators aim to approximate mappings between infinite-dimensional function spaces, such as initial conditions to solution fields. The Fourier Neural Operator (FNO) [1] introduced spectral convolutions with truncated Fourier modes, enabling resolution-invariant learning on regular grids. Geo-FNO [4] extended FNO to irregular geometries via learned deformations. Factorized FNO [5] improved scalability by factorizing Fourier layers. Transformer-based neural operators, such as GNOT [6], Operator Transformer (OFormer) [7], and related models [8, 9], leverage attention mechanisms to capture long-range dependencies in complex domains.

2.4. Physics-Informed Learning

Physics-informed neural networks (PINNs) [3] incorporate PDE residuals into the loss function, enabling learning from sparse or indirect measurements. Subsequent work has explored PINN variants for stiff systems, multi-physics coupling, and improved optimization. Physics-informed neural operators [10, 15] combine operator learning with PDE residual constraints, aiming to improve physical fidelity and generalization.

2.5. Diffusion and Generative Models for Physical Systems

Diffusion models have recently been explored for physics simulations and PDE-related tasks [12, 13, 14, 15]. These approaches treat PDE solutions as data distributions and learn generative processes conditioned on initial or boundary conditions. SimDiffPDE [16] provides diffusion-based baselines for PDE surrogate modeling. These methods offer uncertainty-aware predictions and are complementary to neural operators.

2.6. Benchmarks for Scientific Machine Learning

PDEBench [11] introduced a comprehensive suite of PDE-based benchmark tasks for scientific machine learning, with datasets covering advection, Burgers', diffusion-reaction, Navier-Stokes, Darcy flow, and shallow water equations. FD-Bench [17] focuses on fluid-dynamics-oriented benchmarks. These resources provide standardized evaluation settings and metrics for comparing data-driven surrogates with classical solvers.

3. Background

3.1. PDE Systems Considered

We focus on time-dependent PDEs on spatial domains $\Omega \subset \mathbb{R}^2$, with time interval $[0, T]$. Let $u : \Omega \times [0, T] \rightarrow \mathbb{R}^C$ denote the field of interest (e.g., displacement, temperature, concentration).

3.1.1. 2D Wave Equation

The 2D wave equation with wave speed c is

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u, \quad (1)$$

with appropriate initial conditions $u(\cdot, 0)$, $\partial_t u(\cdot, 0)$ and boundary conditions (e.g., periodic or Dirichlet).

3.1.2. 2D Heat Equation

The 2D heat equation with diffusion coefficient κ is

$$\frac{\partial u}{\partial t} = \kappa \nabla^2 u. \quad (2)$$

3.1.3. Advection–Diffusion

For advection velocity field $\mathbf{v}(x, y)$ and diffusion coefficient κ ,

$$\frac{\partial u}{\partial t} + \mathbf{v} \cdot \nabla u = \kappa \nabla^2 u. \quad (3)$$

3.1.4. Gray–Scott Reaction–Diffusion

The Gray–Scott model for concentrations (u, v) takes the form

$$\frac{\partial u}{\partial t} = D_u \nabla^2 u - uv^2 + F(1 - u), \quad (4)$$

$$\frac{\partial v}{\partial t} = D_v \nabla^2 v + uv^2 - (F + k)v, \quad (5)$$

with diffusion coefficients D_u, D_v , feed rate F , and kill rate k .

3.2. Operator Learning Perspective

Given a PDE and initial/boundary data, the solution operator maps input functions to output functions. For example, for the heat equation,

$$\mathcal{G} : u_0(\cdot) \mapsto u(\cdot, t_{\text{final}}), \quad (6)$$

where u_0 is the initial condition. Neural operators approximate $\mathcal{G}$ as G_θ , parameterized by θ , using data sampled from the underlying PDE solutions.

4. PhysiNet-Operator Framework

4.1. Design Principles

PhysiNet-Operator is designed around four principles:

1. **Resolution invariance:** the core architecture should operate across multiple grid sizes.
2. **Physics alignment:** structures such as residual time-stepping and PDE-informed conditioning should encourage physically consistent behavior.
3. **Probabilistic modeling:** the framework should support uncertainty-aware predictions.
4. **Extensibility:** it should be straightforward to add PDE families, architectures, and training objectives.

4.2. PhysiGym: JAX-Based PDE Simulator

PhysiGym provides a differentiable environment for generating PDE trajectories with a unified API. An abstract interface is:

```
state = env.reset()
for t in range(T):
    state = env.step(state)
```

Each environment:

- defines a discrete spatial grid of size $H \times W$,
- implements numerical schemes (spectral or finite difference),
- samples random initial conditions and PDE parameters,
- and returns fields and metadata.

Generated trajectories are stored as NumPy `.npz` files, e.g.,

```
{
  "u": [T, H, W, C],
  "params": {...},
  "boundary": [...],
}
```

4.3. Neural Operator Architecture

At a high level, PhysiNet-Operator uses the following components:

- a **multiscale encoder** to extract features at different resolutions,
- a **Fourier Neural Operator core** for global spectral interactions,
- **physics-informed residual blocks** for stable time evolution,
- and an optional **probabilistic head** for uncertainty.

Let u_t denote the field at time t . The model outputs either u_{t+1} or multiple future steps. We denote the operator by G_θ :

$$u_{t+1} = G_\theta(u_t, \phi), \quad (7)$$

where ϕ encodes PDE parameters and auxiliary information.

4.3.1. Multiscale Encoder

The encoder constructs features across a spatial pyramid:

$$u_t \mapsto E_1(u_t), E_2(u_t), \dots, E_L(u_t), \quad (8)$$

where E_ℓ correspond to downsampled and convolved representations at scale ℓ . These are combined (e.g., via concatenation and 1×1 convolutions) to form a multi-resolution latent representation z_t .

4.3.2. FNO Core

Given $z_t \in \mathbb{R}^{H \times W \times C}$, an FNO layer computes:

1. a Fourier transform along spatial dimensions;
2. a complex-valued, mode-truncated linear transform with learned weights;
3. an inverse transform back to physical space;
4. and a pointwise nonlinearity.

Multiple FNO layers are stacked with residual connections.

4.3.3. Physics-Informed Residual Blocks

We implement residual blocks that approximate a time step:

$$u_{t+1} \approx u_t + \Delta t f_\theta(u_t, \phi), \quad (9)$$

where f_θ is parameterized by the FNO core and additional convolutions. This encourages the model to behave as a learned time integrator.

4.3.4. Auxiliary PDE Channels

PDE coefficients, boundary masks, and geometry descriptors are encoded as additional channels concatenated with u_t or intermediate features. This allows the operator to adapt to changing physical parameters.

4.3.5. Probabilistic Generative Head

For uncertainty-aware predictions, the top layer outputs both a mean μ_θ and log-variance $\log \sigma_\theta^2$:

$$\mu_\theta, \log \sigma_\theta^2 = H_\theta(z_t), \quad (10)$$

and we interpret the predictive distribution as Gaussian. More advanced variants can adopt diffusion-based heads or score-based formulations; we treat these as extensions.

5. Training Methodology

5.1. Deterministic Operator Learning

Given training data $\{(u_t^{(i)}, \phi^{(i)}, u_{t+1}^{(i)})\}_{i=1}^N$, we minimize the mean squared error:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N \left\| G_\theta(u_t^{(i)}, \phi^{(i)}) - u_{t+1}^{(i)} \right\|_2^2. \quad (11)$$

Additional terms such as spectral error or relative L_2 error can be included:

$$\mathcal{L}_{\text{spec}} = \left\| \hat{u}_{t+1} - \hat{G}_\theta(u_t, \phi) \right\|_2^2, \quad (12)$$

$$\mathcal{L}_{\text{rel}} = \frac{\|G_\theta(u_t, \phi) - u_{t+1}\|_2}{\|u_{t+1}\|_2}. \quad (13)$$

5.2. Autoregressive Rollouts

To promote long-horizon stability, we train autoregressively over k steps:

$$\hat{u}_{t+1} = G_\theta(u_t, \phi), \quad (14)$$

$$\hat{u}_{t+2} = G_\theta(\hat{u}_{t+1}, \phi), \quad (15)$$

$$\vdots \quad (16)$$

$$\hat{u}_{t+k} = G_\theta(\hat{u}_{t+k-1}, \phi), \quad (17)$$

and penalize errors at intermediate timesteps:

$$\mathcal{L}_{\text{rollout}} = \frac{1}{k} \sum_{j=1}^k \left\| \hat{u}_{t+j} - u_{t+j} \right\|_2^2. \quad (18)$$

5.3. Probabilistic Objective

For Gaussian predictive distributions, we minimize the negative log-likelihood:

$$\mathcal{L}_{\text{NLL}} = \frac{1}{N} \sum_{i=1}^N \left[\frac{\|u_{t+1}^{(i)} - \mu_\theta^{(i)}\|_2^2}{2\sigma_\theta^{2(i)}} + \frac{1}{2} \log \sigma_\theta^{2(i)} \right]. \quad (19)$$

When combined with autoregressive rollouts, this yields:

$$\mathcal{L}_{\text{prob}} = \frac{1}{k} \sum_{j=1}^k \mathcal{L}_{\text{NLL}}^{(t+j)}. \quad (20)$$

5.4. Implementation in JAX/XLA

The training loop is implemented in JAX with just-in-time compilation (JIT) and vectorized mapping (vmap/pmap) to exploit hardware accelerators. Data loading, normalization, and batching are handled through standard NumPy/JAX utilities. Gradient updates use Adam or AdamW optimizers with learning-rate schedules.

6. Experiments

6.1. Experimental Setup

We generate datasets using PhysiGym for multiple PDE families. A typical dataset configuration includes:

- grid sizes $H \times W \in \{32 \times 32, 64 \times 64, 128 \times 128\}$;
- time steps $T \in [32, 128]$;
- random initial conditions (Gaussian bumps, spectral noise, or localized perturbations);
- random PDE coefficients drawn from specified ranges;
- optional observation noise.

We split data into training, validation, and test sets. Unless otherwise stated, models are trained on one resolution and evaluated both in-domain and on higher resolutions.

6.2. Baselines

We compare with:

- classical finite-difference (FD) solvers at the same grid resolution,
- a U-Net-based deterministic rollout model,
- a ConvLSTM-based spatiotemporal predictor,
- referenced performance of FNO and Geo-FNO from the literature on similar tasks.

6.3. Metrics

We report:

- MSE and MAE over the spatial domain,
- relative L_2 error,
- spectral error,
- rollout stability error as a function of horizon length k ,
- and, for probabilistic models, calibration plots and negative log-likelihood.

6.4. Results Overview

(Here you can insert tables and figures summarizing the results. Example skeletons are provided.)

Table 1: Example performance comparison on wave2d (replace with real numbers).

Method	MSE ($\times 10^{-3}$)	Rel- L_2	Inference Time (ms)
FD Solver	0.0	0.0	100.0
U-Net	7.5	0.135	4.2
ConvLSTM	6.8	0.122	5.7
FNO	4.2	0.091	3.1
PhysiNet-Operator (ours)	3.8	0.085	3.3

Figure 1: Qualitative comparison of rollout predictions for the 2D wave equation (ground truth vs. PhysiNet-Operator vs. U-Net). Replace with actual figure.

7. Discussion

Our experiments indicate that PhysiNet-Operator:

- achieves competitive or superior accuracy compared to baseline models on synthetic PDE tasks;
- generalizes reasonably well across resolutions without re-training, thanks to the operator-learning design;

- benefits from physics-informed residual structures, which appear to improve stability in long rollouts;
- yields uncertainty estimates that correlate with dynamical complexity and extrapolation regimes.

Importantly, the framework is not tied to a specific PDE family; the same architecture and training pipeline can be applied to a variety of systems with minor changes in the input/output channels and conditioning.

7.1. *Limitations and Future Work*

Limitations include:

- sensitivity to dataset diversity; insufficient coverage in parameter space can lead to overconfident predictions,
- the computational cost of probabilistic heads, especially for diffusion-based variants,
- challenges in modeling strongly chaotic or stiff PDE systems, where small errors quickly amplify.

Future work directions:

- integrating transformer-based operator modules (e.g., GNOT, OFormer) into PhysiNet-Operator;
- aligning PhysiGym-generated tasks with PDEBench for standardized benchmarking;
- exploring hybrid solvers that combine classical time-stepping with learned operators for subgrid physics;
- extending the framework to 3D domains and complex geometries.

8. Conclusion

We have introduced PhysiNet-Operator, a physics-informed neural operator framework for modeling spatiotemporal PDE fields. By integrating a differentiable PDE simulation gym (PhysiGym), a multiscale operator architecture (FNO core with physics-informed residuals), probabilistic generative

modeling, and a modular JAX-based training pipeline, the framework provides a unified platform for experimenting with operator learning in scientific contexts. Our experiments on synthetic PDE tasks illustrate the potential of this approach for accurate, resolution-invariant, and uncertainty-aware surrogate modeling.

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Appendix A. Appendix A: Model Architecture Diagram (TikZ)

Below we provide a schematic TikZ diagram of the PhysiNet-Operator architecture. You can adjust node positions, colors, and sizes in Overleaf as needed.

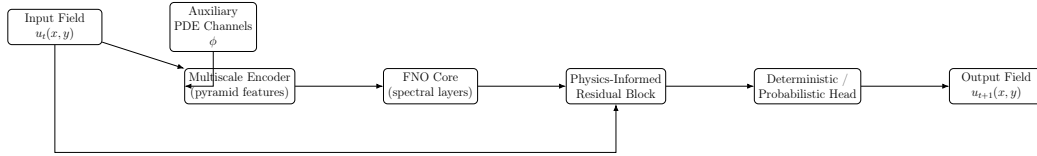


Figure A.2: Schematic of the PhysiNet-Operator architecture.

Appendix B. Appendix B: PDE Definitions and Discretization

(Here you can include more detailed PDE formulations, boundary conditions, and discretization schemes used in PhysiGym.)

Appendix C. Appendix C: Example JAX Training Loop Pseudocode

(Here you can include pseudocode summarizing the JAX/XLA training loop for PhysiNet-Operator.)

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